RENEWABLE ENERGY SOURCES

Previous-Year Numerical Problems

Complete step-by-step solutions

Seven examination papers, 2022–2025 | Final LaTeX edition: 5 November 2026

Scope. All 21 numerical questions in the supplied 20-page compilation are solved in source order. The original wording and data are retained and re-typeset; only spacing and mathematical typography are regularized. Theory questions are excluded. Apparent printing errors and missing data are identified beside the relevant working.

Reading conventions. Boxed, numbered equations give the governing formulas. Working shows substitutions, units and final results. Solar calculations use degree mode and a stated engineering approximation. NPV inflows occur at year ends; money is rounded to two decimals using unrounded discount factors. Biogas dilution assumptions are explicit.

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Source PDF page numbers refer to the supplied compilation, including its blank page 12. The final page provides a results checklist and calculation references.

1 Sunset hour angle and day length

07 February 2023 Q3(b) 10 marks Source PDF p. 1

Original question.

Determine the sunset hour angle and day length at a location latitude of 32° on March 31. The standard longitude for IST is 81° and 44′ E.

Worked solution

1. Day number and latitude.

March 31 in a non-leap year gives $n = 31 + 28 + 31 = 90$. Thus $n = 90$ and $\phi = 32°$.

$$\boxed{\delta = 23.45° \sin\left[\frac{360°(284+n)}{365}\right], \qquad \omega_s = \cos^{-1}(-\tan\phi\tan\delta), \qquad N = \frac{2\omega_s}{15} \quad \text{hours.}} \tag{1.1}$$

Here δ is declination and ω_s is the positive sunset hour angle. All trigonometric calculations use degree mode.

2. Declination.

$$\begin{aligned}\delta &= 23.45° \sin\left[\frac{360°(284+90)}{365}\right] \\ &= 3.6185°.\end{aligned}$$

3. Sunset hour angle.

$$\begin{aligned}\cos\omega_s &= -\tan(32°)\tan(3.6185°) = -0.039516, \\ \omega_s &= \cos^{-1}(-0.039516) = 92.2647°.\end{aligned}$$

4. Daylight duration.

$$\begin{aligned}N &= \frac{2(92.2647)}{15} = 12.3020\,\text{h} \\ &= 12\,\text{h}\ 18.12\,\text{min}.\end{aligned}$$

Final answer. $\omega_s = 92.26°$, $N = 12.302$ h (approximately 12 h 18 min).

Assumption / data note. No year is given, so a non-leap year is assumed. Longitude is unnecessary for sunset hour angle or day length. The result is geometric daylight, without refraction or terrain corrections.

2 Wind power and electrical output

07 February 2023 Q5(a) 10 marks Source PDF p. 2

Original question.

A Horizontal axis wind turbine has following data:

Speed of free wind at a height of 10 m = 13 m/s
Air density = 1.226 kg/m^3
$\alpha = 0.15$
Height of tower = 100 m
Diameter of rotor = 80 m
Wind velocity at the turbine reduces by 20%
Generator efficiency = 85%

Find:

(i) Total power available in wind
(ii) Power extracted by the turbine
(iii) Electrical power generated

Worked solution

Assumption / data note. The 20% reduction is interpreted at the rotor plane: $V_d = 0.80V_1$, hence $a = 0.20$. Ideal actuator-disc theory is used; unspecified drivetrain losses are neglected.

$$\boxed{\begin{aligned} &V_1 = V_{\text{ref}}\left(\frac{z}{z_{\text{ref}}}\right)^{\alpha}, \quad A = \frac{\pi D^2}{4}, \quad P_w = \frac{1}{2}\rho A V_1^3, \\ &C_p = 4a(1-a)^2, \qquad P_t = C_p P_w, \qquad P_e = \eta_g P_t. \end{aligned}} \tag{2.1}$$

1. Hub-height speed and swept area.

$$V_1 = 13\left(\frac{100}{10}\right)^{0.15} = 18.3630\,\text{m s}^{-1},$$
$$A = \frac{\pi(80)^2}{4} = 5\,026.548\,\text{m}^2.$$

2. Available wind power.

$$\begin{aligned} P_w &= \tfrac{1}{2}(1.226)(5026.548)(18.3630)^3 \\ &= 19.079210\,\text{MW}. \end{aligned}$$

3. Extracted power and generator output.

$$\begin{aligned} C_p &= 4(0.20)(0.80)^2 = 0.5120, \\ P_t &= 0.5120(19.079210) = 9.768555\,\text{MW}, \\ P_e &= 0.85(9.768555) = 8.303272\,\text{MW}. \end{aligned}$$

Final answer. (i) $P_w = 19.079\,\text{MW}$; (ii) $P_t = 9.769\,\text{MW}$; (iii) $P_e = 8.303\,\text{MW}$.

Assumption / data note. If the 20% describes the far-wake reduction instead, $a = 0.10$ and $C_p = 0.324$. That interpretation gives $P_t = 6.182\,\text{MW}$ and $P_e = 5.254\,\text{MW}$. The velocity location must be stated.

3 Beam incidence on a south-facing collector

10 December 2024 Q2(a) 10 marks Source PDF p. 3

Original question.

Calculate the angle made by beam radiation with the normal to a flat plate collector on March 15, at 10 A.M. solar time for a location at 35° 45′ N. The collector is tilted at an angle of 45° with the horizontal and is pointing due south.

Worked solution

1. Day number, latitude and solar time.

March 15 gives $n = 31+28+15 = 74$. Convert $35°45' = 35+45/60 = 35.75°$. Thus $n = 74$, $\phi = 35.750000°$, $\beta = 45°$ and $t_s = 10\,\text{h}$.

$$\boxed{\begin{gathered}\delta = 23.45° \sin\left[\frac{360°(284+n)}{365}\right], \qquad \omega = 15°(t_s - 12),\\ \cos\theta = \sin\delta\sin(\phi-\beta) + \cos\delta\cos(\phi-\beta)\cos\omega.\end{gathered}} \tag{3.1}$$

This incidence equation applies to a due-south-facing surface. The angle θ is measured from the collector normal.

2. Declination and hour angle.

$$\delta = 23.45° \sin\left[\frac{360°(284+74)}{365}\right] = -2.8189°,$$
$$\omega = 15°(10-12) = -30°, \qquad \phi - \beta = -9.250000°.$$

3. Substitute and evaluate the incidence angle.

$$\begin{aligned}\cos\theta &= \sin(-2.8189°)\sin(-9.250000°)\\ &\quad + \cos(-2.8189°)\cos(-9.250000°)\cos(-30°)\\ &= 0.861635,\\ \theta &= \cos^{-1}(0.861635) = 30.4994°.\end{aligned}$$

Final answer. The angle made by the beam with the collector normal is $\theta = 30.50°$.

Assumption / data note. The given time is already solar time, so no longitude or equation-of-time correction is needed. A non-leap year is assumed because the year is unspecified.

4 Five-year NPV comparison at 6%

10 December 2024 Q2(b) 10 marks Source PDF p. 4

Original question.

Evaluate the financial viability of two proposed solar power projects using the NPV method. Both projects have the same capital cost but different cash flows over time. The discount rate is 6% for both projects.

	Project A	Project B
Capital cost (Rs.)	45,000	45,000
Year	Net annual saving (Rs.)	Net annual saving (Rs.)
1	10,000	9,500
2	9,500	9,300
3	9,800	9,000
4	9,500	9,200
5	9,000	9,000

i. Calculate the NPV for both projects.
ii. Determine which project is more financially viable based on the NPV values.

Worked solution

Capital cost is paid at $t = 0$; annual savings arrive at year ends. No extra terminal receipt is assumed.

$$\boxed{PV_t = \frac{C_t}{(1+i)^t}, \qquad \text{NPV} = -C_0 + \sum_{t=1}^{n} \frac{C_t}{(1+i)^t}.} \tag{4.1}$$

Here C_0 is initial cost, C_t the annual saving, i the discount rate and n the listed duration.

1. Discount the annual cash flows.

Use $C_0 = \text{Rs. } 45\,000$, $i_A = 6\%$ and $i_B = 6\%$. For year 1, $PV_{A,1} = 10\,000/(1 + 0.06) = \text{Rs. } 9\,433.96$.

Year	PV_A (Rs.)	PV_B (Rs.)
1	9 433.96	8 962.26
2	8 454.97	8 276.97
3	8 228.27	7 556.57
4	7 524.89	7 287.26
5	6 725.32	6 725.32
Total PV	**40 367.41**	**38 808.39**

2. Subtract the initial investment.

$$\text{NPV}_A = 40\,367.41 - 45\,000 = \text{Rs. } -4\,632.59,$$
$$\text{NPV}_B = 38\,808.39 - 45\,000 = \text{Rs. } -6\,191.61.$$

3. Apply the NPV criterion.

Both projects have negative NPV, so neither is financially acceptable at the stated discount rate. Project A has the higher NPV by Rs. 1 559.02.

Final answer. $\text{NPV}_A = \text{Rs. } -4\,632.59$; $\text{NPV}_B = \text{Rs. } -6\,191.61$. Reject both; Project A is merely the less unfavourable option.

5 Three-blade HAWT: power and axial force

10 December 2024 Q5(a) 10 marks Source PDF p. 5

Original question.

The following data were collected for a three-blade Horizontal Axis Wind Turbine (HAWT):

Average wind speed at a standard height of 10 m = 12 m/s
Wind shear coefficient $\alpha = 0.14$
Air density = 1.225 kg/m^3
Hub height = 90 m
Rotor diameter = 70 m
The downstream wind velocity is 60% of the upstream wind velocity.

Estimate the power available in the wind, power extracted by the wind turbine, and axial force acting on the wind turbine.

Worked solution

$$\boxed{\begin{aligned} V_1 &= V_{\text{ref}}\left(\frac{z}{z_{\text{ref}}}\right)^{\alpha}, \quad A = \frac{\pi D^2}{4}, \quad V_2 = kV_1, \\ a &= \frac{1-k}{2}, \quad V_d = \frac{V_1+V_2}{2}, \quad C_p = 4a(1-a)^2, \\ P_w &= \frac{1}{2}\rho A V_1^3, \quad P_t = C_p P_w, \quad F = \frac{1}{2}\rho A V_1^2(1-k^2). \end{aligned}} \tag{5.1}$$

1. Hub-height speed and swept area.

$$V_1 = 12\left(\frac{90}{10}\right)^{0.14} = 16.3221\,\text{m s}^{-1},$$
$$A = \frac{\pi(70)^2}{4} = 3\,848.451\,\text{m}^2.$$

2. Downstream speed, rotor-plane speed and induction.

$$V_2 = 0.6(16.3221) = 9.7932\,\text{m s}^{-1},$$
$$a = \frac{1-0.6}{2} = 0.2, \qquad V_d = \frac{V_1+V_2}{2} = 13.0577\,\text{m s}^{-1},$$
$$C_p = 4(0.2)(1-0.2)^2 = 0.5120.$$

3. Available and extracted power.

$$P_w = \tfrac{1}{2}(1.225)(3848.451)(16.3221)^3 = 10.249853\,\text{MW},$$
$$P_t = 0.5120(10.249853) = 5.247925\,\text{MW}.$$

4. Axial force.

$$F = \tfrac{1}{2}(1.225)(3848.451)(16.3221)^2[1-(0.6)^2]$$
$$= 401\,904.09\,\text{N} = 401.904\,\text{kN}.$$

Final answer. $P_w = 10.250\,\text{MW}$; $P_t = 5.248\,\text{MW}$; $F = 401.904\,\text{kN}$.

Assumption / data note. The given percentage is a far-downstream velocity ratio. Rotor-plane speed is the mean of upstream and downstream speeds. Blade count does not enter this ideal actuator-disc model.

6 Kanpur: sunset angle in a leap year

16 February 2024 Q3(a) 8 marks Source PDF p. 6

Original question.

Estimate Sunset hour angle, and daylight hours (sunshine hours) at Kanpur (latitude 26.45° North) on June 28th in a leap year.

Worked solution

1. Day number and latitude.

June 28 in a leap year gives $n = 31 + 29 + 31 + 30 + 31 + 28 = 180$. Thus $n = 180$ and $\phi = 26.45°$.

$$\boxed{\delta = 23.45° \sin\left[\frac{360°(284+n)}{365}\right], \qquad \omega_s = \cos^{-1}(-\tan\phi\tan\delta), \qquad N = \frac{2\omega_s}{15} \quad \text{hours.}} \tag{6.1}$$

Here δ is declination and ω_s is the positive sunset hour angle. All trigonometric calculations use degree mode.

2. Declination.

$$\begin{aligned}\delta &= 23.45° \sin\left[\frac{360°(284+180)}{365}\right] \\ &= 23.2416°.\end{aligned}$$

3. Sunset hour angle.

$$\begin{aligned}\cos\omega_s &= -\tan(26.45°)\tan(23.2416°) = -0.213654, \\ \omega_s &= \cos^{-1}(-0.213654) = 102.3365°.\end{aligned}$$

4. Daylight duration.

$$\begin{aligned}N &= \frac{2(102.3365)}{15} = 13.6449\,\text{h} \\ &= 13\,\text{h}\ 38.69\,\text{min}.\end{aligned}$$

Final answer. $\omega_s = 102.34°$, $N = 13.645$ h (approximately 13 h 39 min).

Assumption / data note. The conventional 365-day Cooper approximation is used with the actual leap-year ordinal $n = 180$. This is an engineering estimate of geometric daylight.

7 VAWT: speed, induction and conditional output

16 February 2024 Q4(a) 8 marks Source PDF p. 7

Original question.

A wind turbine (VAWT) is to be installed at a tower of height 120 m in a locality. Considering the given data for the system Calculate

i. Speed of wind at tower height
ii. Total power available in wind
iii. Interference factor
iv. Power coefficient
v. Power extracted by the turbine
vi. Electrical power generated by the generator

Given:

1. Speed of the free wind at a height of 10 m = 15 m/s
2. Air density = 1.226 kg/m^3
3. α (constant) = 0.14
4. Diameter of rotor = 100 m
5. wind velocity at the turbine reduces by 25%
6. Generator efficiency = 90%

Worked solution

Assumption / data note. A true VAWT has swept area $A = DH$; blade height H is missing. Actual powers therefore depend on H. The rotor-plane reduction gives $a = 0.25$. A circular-area interpretation is evaluated separately.

$$\boxed{\begin{aligned} V_1 &= V_{\text{ref}}\left(\frac{z}{z_{\text{ref}}}\right)^{\alpha}, \quad a = 1 - \frac{V_d}{V_1}, \quad C_p = 4a(1-a)^2, \\ P_w &= \frac{1}{2}\rho A V_1^3, \qquad P_t = C_p P_w, \qquad P_e = \eta_g P_t. \end{aligned}} \tag{7.1}$$

1. Speed, interference factor and power coefficient.

$$\begin{aligned} V_1 &= 15(120/10)^{0.14} = 21.2411\,\text{m s}^{-1}, \\ a &= 1 - 0.75 = 0.25, \qquad C_p = 4(0.25)(0.75)^2 = 0.5625. \end{aligned}$$

2. Powers for the stated VAWT geometry.

Let H be blade height in metres. Then $A = 100H\,\text{m}^2$ and

$$\begin{aligned} P_w &= \tfrac{1}{2}(1.226)(100H)(21.2411)^3 = 587.476H\,\text{kW}, \\ P_t &= 0.5625P_w = 330.455H\,\text{kW}, \\ P_e &= 0.90P_t = 297.410H\,\text{kW}. \end{aligned}$$

3. Conditional circular-area calculation.

If the examiner intended a circular swept area, assume $A = \pi D^2/4$:

$$\begin{aligned} A &= \frac{\pi(100)^2}{4} = 7\,853.982\,\text{m}^2, \quad P_w = 46.140283\,\text{MW}, \\ P_t &= 0.5625P_w = 25.953909\,\text{MW}, \\ P_e &= 0.90P_t = 23.358518\,\text{MW}. \end{aligned}$$

Final answer. $V_1 = 21.241\,\text{m s}^{-1}$, $a = 0.25$, $C_p = 0.5625$. Under the circular-area assumption only, $(P_w, P_t, P_e) = (46.140,\ 25.954,\ 23.359)\,\text{MW}$. Actual VAWT powers depend on H.

8 Biogas demand, required cows and digester volume

16 February 2024 Q4(b) 8 marks Source PDF p. 7

Original question.

A family of 4 adults plan to install a biogas plant for their cooking and lighting needs with seven 100 CP lamps for six hours daily. Calculate:

(i) Number of cows required to feed the biogas plant
(ii) Volume of the digester

Consider the data given below:

- Cow dung production rate is 15 kg /day/head with Gas yield of 0.40 m^3 per kg of dry matter.
- Only 60% of the cow dung produced is collectible.
- Solid (dry) matter content in cow dung is about 18% only.
- Biogas required for cooking is about 0.35 m^3/person/day
- Gas required for lighting of 100 CP (candle Power) mantle lamp is 0.15 m^3/hour.
- The density of slurry is about 1090 kg/m^3.
- Consider 40-day retention time for volume of slurry in the digester and 90% occupied.
- Heating value of biogas = 23MJ/m^3

Worked solution

Assumption / data note. Added water is unspecified. Assume fresh dung : added water = 1:1 by mass, slurry density $\rho_s = 1090\,\mathrm{kg\,m^{-3}}$, and liquid occupancy $f_o = 0.90$. HRT denotes hydraulic retention time. Total digester volume differs from daily gas output.

$$\boxed{\begin{gathered} G = N_p g_c + N_L t_L g_L, \quad G_{\text{cow}} = m_{\text{dung,cow}} f_c f_d Y, \quad N_{\text{cow}} = \left\lceil \frac{G}{G_{\text{cow}}} \right\rceil, \\ m_d = \frac{G}{Y}, \quad m_w = \frac{m_d}{f_d}, \quad V_{\text{dig}} = \frac{(m_w + m_{\text{water}})\,\text{HRT}}{\rho_s f_o}. \end{gathered}} \tag{8.1}$$

G is daily gas demand; f_c is collectible fraction; $f_d = 0.18$ is dry-matter fraction; Y is gas yield per kg of dry matter; m_w is fresh-dung feed.

1. Daily cooking and lighting demand.

$$G_c = 4(0.35) = 1.400\,\mathrm{m^3\,day^{-1}},$$
$$G_L = 7(6)(0.15) = 6.300\,\mathrm{m^3\,day^{-1}},$$
$$G = 1.400 + 6.300 = 7.700\,\mathrm{m^3\,day^{-1}}.$$

2. Gas yield per cow and required herd.

$$m_{w,\mathrm{cow}} = 15(0.6) = 9.000\,\mathrm{kg\,day^{-1}},$$
$$m_{d,\mathrm{cow}} = 9.000(0.18) = 1.620\,\mathrm{kg\,day^{-1}},$$
$$G_{\mathrm{cow}} = 1.620(0.4) = 0.6480\,\mathrm{m^3\,day^{-1}},$$
$$N_{\mathrm{cow}} = \left\lceil \frac{7.700}{0.6480} \right\rceil = \lceil 11.8827 \rceil = 12.$$

3. Minimum fresh-dung and slurry feed.

$$m_d = \frac{7.700}{0.4} = 19.2500\,\mathrm{kg\,day^{-1}},$$
$$m_w = \frac{19.2500}{0.18} = 106.9444\,\mathrm{kg\,day^{-1}},$$
$$m_s = m_w + m_{\mathrm{water}} = 2(106.9444) = 213.8889\,\mathrm{kg\,day^{-1}}.$$

4. Demand-based digester volume.

$$\dot{V}_s = \frac{213.8889}{1090} = 0.196228\,\mathrm{m^3\,day^{-1}},$$
$$V_{\mathrm{dig}} = \frac{(0.196228)(40)}{0.90} = 8.7213\,\mathrm{m^3}.$$

5. Volume if all collectible dung from the rounded herd is fed.

$$m_{w,\mathrm{herd}} = 12(15)(0.6) = 108.000\,\mathrm{kg\,day^{-1}},$$
$$V_{\mathrm{herd}} = \frac{2(108.000)(40)}{1090(0.90)} = 8.8073\,\mathrm{m^3}.$$

Final answer. Required herd: 12 cows. Minimum demand-based volume: $8.72\,\mathrm{m^3}$. For all collectible dung from the 12 cows, provide $8.81\,\mathrm{m^3}$.

Assumption / data note. The heating value is not required for the requested sizing. Without added dilution water, the computed volumes would be half these values; that is a different feed assumption.

9 Ten-year NPV comparison at 6%

16 February 2024 Q5(b) 10 marks Source PDF p. 8

Original question.

Two proposed projects are presented in the table given below:

	Project 1	Project 2
Capital cost (Rs.)	50,000	50,000
Year	Net annual saving (Rs.)	Net annual saving (Rs.)
1	+8,000	+8,600
2	+8,000	+8,600
3	+8,000	+8,300
4	+8,000	+8,300
5	+8,000	+8,000
6	+8,000	+8,000
7	+8,000	+7,700
8	+8,000	+7,700
9	+8,000	+7,400
10	+8,000	+7,400
Total net savings at end of 10th year	80,000	80,000

Evaluate the financial viability of two proposed projects by using the Net Present Value (NPV) analysis technique. Both projects are subject to an annual discount rate of 6%.

Worked solution

Capital cost is paid at $t = 0$; annual savings arrive at year ends. No extra terminal receipt is assumed.

$$\boxed{PV_t = \frac{C_t}{(1+i)^t}, \qquad \text{NPV} = -C_0 + \sum_{t=1}^{n} \frac{C_t}{(1+i)^t}.} \tag{9.1}$$

Here C_0 is initial cost, C_t the annual saving, i the discount rate and n the listed duration.

1. Discount the annual cash flows.

Use $C_0 = \text{Rs. } 50\,000$, $i_1 = 6\%$ and $i_2 = 6\%$. For year 1, $PV_{1,1} = 8\,000/(1 + 0.06) = \text{Rs. } 7\,547.17$.

Year	PV_1 (Rs.)	PV_2 (Rs.)
1	7 547.17	8 113.21
2	7 119.97	7 653.97
3	6 716.95	6 968.84
4	6 336.75	6 574.38
5	5 978.07	5 978.07
6	5 639.68	5 639.68
7	5 320.46	5 120.94
8	5 019.30	4 831.08
9	4 735.19	4 380.05
10	4 467.16	4 132.12
Total PV	**58 880.70**	**59 392.33**

2. Subtract the initial investment.

$$\text{NPV}_1 = 58\,880.70 - 50\,000 = \text{Rs. } 8\,880.70,$$
$$\text{NPV}_2 = 59\,392.33 - 50\,000 = \text{Rs. } 9\,392.33.$$

3. Apply the NPV criterion.

Both projects have positive NPV and are acceptable at their stated discount rates. Project 2 has the higher NPV by Rs. 511.63.

Final answer. $\text{NPV}_1 = \text{Rs. } 8\,880.70$; $\text{NPV}_2 = \text{Rs. } 9\,392.33$. Select Project 2.

Assumption / data note. The Rs. 80,000 total is the sum of the ten annual savings, not an extra terminal receipt.

10 Beam incidence on a tilted solar collector

25 February 2025 Q2(a) 10 marks Source PDF p. 9

Original question.

Determine the angle made by beam radiation with the normal to a solar collector on July 25, at 11 A.M. solar time for a location at 40° 20′ N. The solar collector is inclined at an angle of 30° with respect to the horizontal and is facing due south.

Worked solution

1. Day number, latitude and solar time.

July 25 gives $n = 31 + 28 + 31 + 30 + 31 + 30 + 25 = 206$. Convert $40°20' = 40 + 20/60 = 40.333333°$. Thus $n = 206$, $\phi = 40.333333°$, $\beta = 30°$ and $t_s = 11\,\text{h}$.

$$\boxed{\begin{gathered}\delta = 23.45° \sin\left[\frac{360°(284+n)}{365}\right], \qquad \omega = 15°(t_s - 12), \\ \cos\theta = \sin\delta\sin(\phi-\beta) + \cos\delta\cos(\phi-\beta)\cos\omega.\end{gathered}} \tag{10.1}$$

This incidence equation applies to a due-south-facing surface. The angle θ is measured from the collector normal.

2. Declination and hour angle.

$$\delta = 23.45° \sin\left[\frac{360°(284+206)}{365}\right] = 19.6025°,$$
$$\omega = 15°(11-12) = -15°, \qquad \phi - \beta = 10.333333°.$$

3. Substitute and evaluate the incidence angle.

$$\begin{aligned}\cos\theta &= \sin(19.6025°)\sin(10.333333°) \\ &\quad + \cos(19.6025°)\cos(10.333333°)\cos(-15°) \\ &= 0.955364, \\ \theta &= \cos^{-1}(0.955364) = 17.1834°.\end{aligned}$$

Final answer. The angle made by the beam with the collector normal is $\theta = 17.18°$.

Assumption / data note. The given time is already solar time, so no longitude or equation-of-time correction is needed. A non-leap year is assumed because the year is unspecified.

11 Six-year NPV comparison at 5%

25 February 2025 Q2(b) 10 marks Source PDF p. 10

Original question.

Evaluate the financial viability of two proposed solar power projects using the NPV method. Both projects have the same capital cost but different cash flows over time. The discount rate is 5% for both projects.

	Project A	Project B
Capital cost (Rs.)	40,000	40,000
Year	Net annual saving (Rs.)	Net annual saving (Rs.)
1	8900	8600
2	8000	8600
3	8900	8300
4	8000	8300
5	8000	8000
6	8000	8000

i. Calculate the NPV for both projects.
ii. Determine which project is more financially viable based on the NPV values.

Worked solution

Capital cost is paid at $t = 0$; annual savings arrive at year ends. No extra terminal receipt is assumed.

$$\boxed{PV_t = \frac{C_t}{(1+i)^t}, \qquad \text{NPV} = -C_0 + \sum_{t=1}^{n} \frac{C_t}{(1+i)^t}.} \tag{11.1}$$

Here C_0 is initial cost, C_t the annual saving, i the discount rate and n the listed duration.

1. Discount the annual cash flows.

Use $C_0 = \text{Rs. } 40\,000$, $i_A = 5\%$ and $i_B = 5\%$. For year 1, $PV_{A,1} = 8\,900/(1 + 0.05) = \text{Rs. } 8\,476.19$.

Year	PV_A (Rs.)	PV_B (Rs.)
1	8 476.19	8 190.48
2	7 256.24	7 800.45
3	7 688.15	7 169.85
4	6 581.62	6 828.43
5	6 268.21	6 268.21
6	5 969.72	5 969.72
Total PV	**42 240.13**	**42 227.14**

2. Subtract the initial investment.

$$\text{NPV}_A = 42\,240.13 - 40\,000 = \text{Rs. } 2\,240.13,$$
$$\text{NPV}_B = 42\,227.14 - 40\,000 = \text{Rs. } 2\,227.14.$$

3. Apply the NPV criterion.

Both projects have positive NPV and are acceptable at their stated discount rates. Project A has the higher NPV by Rs. 12.99.

Final answer. $\text{NPV}_A = \text{Rs. } 2\,240.13$; $\text{NPV}_B = \text{Rs. } 2\,227.14$. Select Project A.

12 Four-blade HAWT: power and axial force

25 February 2025 Q5(a) 10 marks Source PDF p. 11

Original question.

For a four-blade HAWT, the following observations were made:

Average wind speed at a standard height of 10 m = 10 m/s
Wind shear coefficient $\alpha = 0.15$
Air density = 1.23 kg/m^3
Hub height = 100 m
Rotor diameter = 80 m
The downstream wind velocity is 70% of the upstream wind velocity

i. Compute the power available in the wind.
ii. Determine the power extracted by the wind turbine.
iii. Calculate the axial force exerted on the wind turbine.

Worked solution

$$\boxed{\begin{aligned} V_1 &= V_{\text{ref}}\left(\frac{z}{z_{\text{ref}}}\right)^{\alpha}, \quad A = \frac{\pi D^2}{4}, \quad V_2 = kV_1, \\ a &= \frac{1-k}{2}, \quad V_d = \frac{V_1+V_2}{2}, \quad C_p = 4a(1-a)^2, \\ P_w &= \frac{1}{2}\rho A V_1^3, \quad P_t = C_p P_w, \quad F = \frac{1}{2}\rho A V_1^2(1-k^2). \end{aligned}} \tag{12.1}$$

1. Hub-height speed and swept area.

$$V_1 = 10\left(\frac{100}{10}\right)^{0.15} = 14.1254\,\text{m s}^{-1},$$
$$A = \frac{\pi(80)^2}{4} = 5\,026.548\,\text{m}^2.$$

2. Downstream speed, rotor-plane speed and induction.

$$V_2 = 0.7(14.1254) = 9.8878\,\text{m s}^{-1},$$
$$a = \frac{1-0.7}{2} = 0.15, \qquad V_d = \frac{V_1+V_2}{2} = 12.0066\,\text{m s}^{-1},$$
$$C_p = 4(0.15)(1-0.15)^2 = 0.4335.$$

3. Available and extracted power.

$$P_w = \tfrac{1}{2}(1.23)(5026.548)(14.1254)^3 = 8.712544\,\text{MW},$$
$$P_t = 0.4335(8.712544) = 3.776888\,\text{MW}.$$

4. Axial force.

$$F = \tfrac{1}{2}(1.23)(5026.548)(14.1254)^2[1-(0.7)^2]$$
$$= 314\,568.44\,\text{N} = 314.568\,\text{kN}.$$

Final answer. $P_w = 8.713\,\text{MW}$; $P_t = 3.777\,\text{MW}$; $F = 314.568\,\text{kN}$.

Assumption / data note. The given percentage is a far-downstream velocity ratio. Rotor-plane speed is the mean of upstream and downstream speeds. Blade count does not enter this ideal actuator-disc model.

13 Bhubaneswar: sunset angle and daylight

04 December 2023 Q3(a) 8 marks Source PDF p. 14

Original question.

Estimate the sunset hour angle and the duration of daylight hours (sunshine hours) at Bhubaneswar, which is located at a latitude of 20.29° North, on April 10th in a non-leap year.

Worked solution

1. Day number and latitude.

April 10 in a non-leap year gives $n = 31 + 28 + 31 + 10 = 100$. Thus $n = 100$ and $\phi = 20.29°$.

$$\boxed{\delta = 23.45° \sin\left[\frac{360°(284+n)}{365}\right], \qquad \omega_s = \cos^{-1}(-\tan\phi\tan\delta), \qquad N = \frac{2\omega_s}{15} \quad \text{hours.}} \tag{13.1}$$

Here δ is declination and ω_s is the positive sunset hour angle. All trigonometric calculations use degree mode.

2. Declination.

$$\begin{aligned}\delta &= 23.45° \sin\left[\frac{360°(284+100)}{365}\right] \\ &= 7.5338°.\end{aligned}$$

3. Sunset hour angle.

$$\begin{aligned}\cos\omega_s &= -\tan(20.29°)\tan(7.5338°) = -0.048895, \\ \omega_s &= \cos^{-1}(-0.048895) = 92.8026°.\end{aligned}$$

4. Daylight duration.

$$\begin{aligned}N &= \frac{2(92.8026)}{15} = 12.3737\,\text{h} \\ &= 12\,\text{h}\ 22.42\,\text{min}.\end{aligned}$$

Final answer. $\omega_s = 92.80°$, $N = 12.374\,\text{h}$ (approximately 12 h 22 min).

Assumption / data note. 20.29° is decimal degrees, not 20°29′. Geometric daylight is calculated, without atmospheric refraction.

14 Maximum shaft torque: 100 m rotor

04 December 2023 Q4(a) 8 marks Source PDF p. 14

Original question.

Following data were measured for a HAWT:

Speed of wind = 15 m/s at 1 atm and 27 °C
Diameter of rotor = 100 m
Speed of rotor = 45 rpm

Calculate the torque produced at the shaft for maximum output of the turbine.

Worked solution

Assumption / data note. Maximum possible shaft output is interpreted as the ideal Betz-limit aerodynamic output, with no unspecified drivetrain loss. Actual turbine torque would be lower.

$$\boxed{\begin{aligned} &\rho = \frac{p}{R_{\text{air}}T}, \quad A = \frac{\pi D^2}{4}, \quad P_w = \frac{1}{2}\rho A V^3, \\ &P_{\max} = \frac{16}{27}P_w, \qquad \Omega = \frac{2\pi N}{60}, \qquad \tau_{\max} = \frac{P_{\max}}{\Omega}. \end{aligned}} \tag{14.1}$$

1. Air density at the stated conditions.

Use $p = 1\,\text{atm} = 101\,325\,\text{Pa}$ and $R_{\text{air}} = 287\,\text{J kg}^{-1}\,\text{K}^{-1}$.

$$\begin{aligned} T &= 27 + 273.15 = 300.15\,\text{K}, \\ \rho &= \frac{101\,325}{287(300.15)} = 1.176241\,\text{kg m}^{-3}. \end{aligned}$$

2. Swept area and power in the wind.

$$\begin{aligned} A &= \frac{\pi(100)^2}{4} = 7\,853.982\,\text{m}^2, \\ P_w &= \tfrac{1}{2}(1.176241)(7853.982)(15)^3 \\ &= 15.589423\,\text{MW}. \end{aligned}$$

3. Betz-limit output and angular speed.

$$\begin{aligned} P_{\max} &= \frac{16}{27}(15.589423) = 9.238176\,\text{MW}, \\ \Omega &= \frac{2\pi(45)}{60} = 4.712389\,\text{rad s}^{-1}. \end{aligned}$$

4. Maximum shaft torque.

$$\begin{aligned} \tau_{\max} &= \frac{9.238176 \times 10^6}{4.712389} \\ &= 1\,960\,401.91\,\text{N m}. \end{aligned}$$

Final answer. $\tau_{\max} = 1.9604\,\text{MN m} = 1\,960.40\,\text{kN m}$.

Assumption / data note. Use watts and radians per second in $\tau = P/\Omega$. Rotor rpm cannot be inserted directly as angular speed.

15 Fixed-dome digester volume and thermal power

04 December 2023 Q4(b) 8 marks Source PDF p. 14

Original question.

Calculate the volume of a fixed dome type biogas digester for the output of 10 cows. Also calculate the thermal power available from biogas. Use following data.

Retention time: 40 days,
Dry matter produced: 3kg/day/cow
Biogas yield: 0.30 m^3/kg of dry matter,
% of dry matter in cow dung:18%
Density of slurry:1090 kg/m^3, Burner efficiency:70%
Heating value of biogas: 23MJ/m^3
% volume of the digester occupied by slurry: 90%

Worked solution

Assumption / data note. Assume all stated dry matter is fed and fresh dung : added water = 1:1 by mass. The 70% burner efficiency determines useful thermal output. Both gross gas power and useful burner power are reported.

$$\boxed{\begin{aligned} m_d &= N_{\text{cow}} m_{d,\text{cow}}, \quad m_w = \frac{m_d}{f_d}, \quad V_{\text{dig}} = \frac{(m_w + m_{\text{water}})\,\text{HRT}}{\rho_s f_o}, \\ G &= m_d Y, \qquad P_{\text{gas}} = \frac{G\,HV\,10^6}{86\,400}, \qquad P_{\text{useful}} = \eta_b P_{\text{gas}}. \end{aligned}} \tag{15.1}$$

1. Daily solids, fresh dung and slurry.

$$\begin{aligned} m_d &= 10(3) = 30\,\text{kg day}^{-1}, \\ m_w &= 30/0.18 = 166.6667\,\text{kg day}^{-1}, \\ m_s &= 2m_w = 333.3333\,\text{kg day}^{-1}. \end{aligned}$$

2. Digester volume.

$$\begin{aligned} \dot{V}_s &= 333.3333/1090 = 0.305810\,\text{m}^3\,\text{day}^{-1}, \\ V_{\text{liquid}} &= (0.305810)(40) = 12.2324\,\text{m}^3, \\ V_{\text{dig}} &= 12.2324/0.90 = 13.5916\,\text{m}^3. \end{aligned}$$

3. Gas generation and gross thermal output.

$$\begin{aligned} G &= 30(0.30) = 9.00\,\text{m}^3\,\text{day}^{-1}, \\ E_{\text{day}} &= 9(23) = 207\,\text{MJ day}^{-1}, \\ P_{\text{gas}} &= \frac{207 \times 10^6}{86\,400} = 2395.833\,\text{W} = 2.3958\,\text{kW}. \end{aligned}$$

4. Useful burner output.

$$P_{\text{useful}} = 0.70(2.3958) = 1.6771\,\text{kW}.$$

Final answer. $V_{\text{dig}} = 13.59\,\text{m}^3$; gross thermal power 2.396 kW; useful thermal power 1.677 kW.

Assumption / data note. Power is averaged over 24 hours. With no added water the volume is 6.80 m^3; gas yield and thermal power are unchanged.

16 NPV comparison with a mislabelled total row

04 December 2023 Q5(a) 10 marks Source PDF p. 15

Original question.

Conduct a Net Present Value (NPV) analysis for the two proposed projects presented in the table below, considering an annual discount rate of 5%, to assess their financial viability.

	Project 1	Project 2
Capital cost (Rs.)	50,000	50,000
Year	Net annual saving (Rs.)	Net annual saving (Rs.)
1	+12,000	+15,000
2	+12,000	+15,000
3	+12,000	+12,000
4	+12,000	+12,000
5	+12,000	+6,000
Total net savings at end of 10th year	60,000	60,000

Worked solution

Capital cost is paid at $t = 0$; annual savings arrive at year ends. No extra terminal receipt is assumed.

$$\boxed{PV_t = \frac{C_t}{(1+i)^t}, \qquad \text{NPV} = -C_0 + \sum_{t=1}^{n} \frac{C_t}{(1+i)^t}.} \tag{16.1}$$

Here C_0 is initial cost, C_t the annual saving, i the discount rate and n the listed duration.

1. Discount the annual cash flows.

Use $C_0 = \text{Rs. } 50\,000$, $i_1 = 5\%$ and $i_2 = 5\%$. For year 1, $PV_{1,1} = 12\,000/(1 + 0.05) = \text{Rs. } 11\,428.57$.

Year	PV_1 (Rs.)	PV_2 (Rs.)
1	11 428.57	14 285.71
2	10 884.35	13 605.44
3	10 366.05	10 366.05
4	9 872.43	9 872.43
5	9 402.31	4 701.16
Total PV	**51 953.72**	**52 830.79**

2. Subtract the initial investment.

$$\text{NPV}_1 = 51\,953.72 - 50\,000 = \text{Rs. } 1\,953.72,$$
$$\text{NPV}_2 = 52\,830.79 - 50\,000 = \text{Rs. } 2\,830.79.$$

3. Apply the NPV criterion.

Both projects have positive NPV and are acceptable at their stated discount rates. Project 2 has the higher NPV by Rs. 877.07.

Final answer. $\text{NPV}_1 = \text{Rs. } 1\,953.72$; $\text{NPV}_2 = \text{Rs. } 2\,830.79$. Select Project 2.

Assumption / data note. The total row says "end of 10th year", although only years 1–5 are supplied and their savings sum to Rs. 60,000. The analysis uses these five stated flows. The total is not added as a terminal receipt. A ten-year appraisal would need the missing years 6–10.

17 New Delhi: solar zenith angle from IST

10 December 2025 Q2(a) 10 marks Source PDF p. 16

Original question.

For New Delhi (28° 35′ N, 77° 12′ E), calculate the zenith angle of the sun at 2:30 P.M. on 20 February 2015. The standard IST latitude for India is 81° 44′ E.

Worked solution

Assumption / data note. The stated time is treated as IST. The printed phrase "IST latitude" is interpreted as standard longitude. First use the supplied 81°44′ E; then show the result for the actual IST meridian, 82°30′ E.

$$\boxed{\begin{gathered}\delta = 23.45^\circ \sin\left[\frac{360^\circ(284+n)}{365}\right], \qquad B = \frac{360^\circ(n-81)}{364},\\ E = 9.87\sin 2B - 7.53\cos B - 1.5\sin B \quad \text{(min)},\\ t_s = t_{\text{IST}} + \frac{4(\lambda-\lambda_{\text{st}})+E}{60}, \qquad \omega = 15^\circ(t_s-12),\\ \cos\theta_z = \sin\phi\sin\delta + \cos\phi\cos\delta\cos\omega.\end{gathered}} \tag{17.1}$$

1. Coordinates, day number and declination.

$$\begin{aligned} &n = 31 + 20 = 51, \quad \phi = 28 + 35/60 = 28.583333^\circ,\\ &\lambda = 77 + 12/60 = 77.200000^\circ\,\text{E}, \quad \lambda_{\text{st}} = 81 + 44/60 = 81.733333^\circ\,\text{E},\\ &\delta = 23.45^\circ \sin[360^\circ(284+51)/365] = -11.5790^\circ. \end{aligned}$$

2. Equation of time and local solar time.

$$\begin{aligned} B &= 360^\circ(51-81)/364 = -29.6703^\circ,\\ E &= 9.87\sin 2B - 7.53\cos B - 1.5\sin B = -14.2905\,\text{min},\\ \Delta t &= 4(77.200000 - 81.733333) + (-14.2905)\\ &= -32.4239\,\text{min},\\ t_s &= 14.5 + (-32.4239)/60 = 13.959602\,\text{h},\\ \omega &= 15^\circ(13.959602 - 12) = 29.3940^\circ. \end{aligned}$$

3. Solar zenith angle.

$$\begin{aligned} \cos\theta_z &= \sin(28.583333^\circ)\sin(-11.5790^\circ)\\ &\quad + \cos(28.583333^\circ)\cos(-11.5790^\circ)\cos(29.3940^\circ)\\ &= 0.653475, \quad \theta_z = 49.1959^\circ. \end{aligned}$$

Final answer. Using the printed meridian, $\theta_z = 49.20^\circ$. Using the actual IST meridian 82°30′ E, $\theta_z = 48.77^\circ$.

Assumption / data note. For the actual IST meridian, $t_s = 13.908491$ h and $\omega = 28.6274^\circ$. Declination and equation of time use the explicitly stated examination-level approximations.

18 NPV comparison at different discount rates

10 December 2025 Q2(b) 10 marks Source PDF p. 17

Original question.

Compare projects A and B using the given data. Use the Net Present Value (NPV) method of evaluation.

	Project A	Project B
Capital cost (Rs.)	10,00,000	10,00,000
Cost of capital	15%	13%
Year	Net annual saving (Rs.)	Net annual saving (Rs.)
1	2,00,000	3,00,000
2	3,00,000	4,00,000
3	4,00,000	4,00,000
4	3,00,000	3,00,000
5	1,00,000	2,00,000

i. Calculate the NPV for both projects.
ii. Determine which project is more financially viable based on the NPV values.

Worked solution

Capital cost is paid at $t = 0$; annual savings arrive at year ends. No extra terminal receipt is assumed.

$$PV_t = \frac{C_t}{(1+i)^t}, \qquad \text{NPV} = -C_0 + \sum_{t=1}^{n} \frac{C_t}{(1+i)^t}. \tag{18.1}$$

Here C_0 is initial cost, C_t the annual saving, i the discount rate and n the listed duration.

1. Discount the annual cash flows.

Use $C_0 = \text{Rs. } 1\,000\,000$, $i_A = 15\%$ and $i_B = 13\%$. For year 1, $PV_{A,1} = 200\,000/(1+0.15) = \text{Rs. } 173\,913.04$.

Year	PV_A (Rs.)	PV_B (Rs.)
1	173 913.04	265 486.73
2	226 843.10	313 258.67
3	263 006.49	277 220.06
4	171 525.97	183 995.62
5	49 717.67	108 551.99
Total PV	**885 006.28**	**1 148 513.07**

2. Subtract the initial investment.

$$\text{NPV}_A = 885\,006.28 - 1\,000\,000 = \text{Rs. } -114\,993.72,$$
$$\text{NPV}_B = 1\,148\,513.07 - 1\,000\,000 = \text{Rs. } 148\,513.07.$$

3. Apply the NPV criterion.

Only Project B has positive NPV and is financially acceptable at its stated discount rate. Project B has the higher NPV by Rs. 263 506.79.

Final answer. $\text{NPV}_A = \text{Rs. } -114\,993.72$; $\text{NPV}_B = \text{Rs. } 148\,513.07$. Select Project B.

Assumption / data note. Use the project-specific costs of capital: 15% for A and 13% for B. The printed capital cost 10,00,000 means Rs. 1,000,000 (ten lakh).

19 Family biogas plant: five adults

10 December 2025 Q3(a) 10 marks Source PDF p. 17

Original question.

Calculate the volume of a cow dung-based biogas plant required for cooking needs of a family of five adults and lighting need with two 100 C.P. lamps for three hours daily. Also calculate the required number of cows to feed the plant. Assume standard values of data where required.

Following data may be considered for BIOMASS Numerical:

- Cow dung production rate is 10 kg /day/head with Gas yield of 0.34 m^3 per kg of dry matter.
- Only 70% of the cow dung produced is collectible.
- Solid matter content in cow dung is about 18% only.
- Biogas required for cooking is about 0.227 m^3/person/day
- Gas required for lighting of 100 CP (candle Power) mantle lamp is 0.126 m^3/hour.
- The density of slurry is about 1090 kg/m^3.
- Consider 50-day retention time for volume of slurry in the digester and 90% occupied.
- Heating value of biogas = 23MJ/m^3

Worked solution

Assumption / data note. Added water is unspecified. Assume fresh dung : added water = 1:1 by mass, slurry density $\rho_s = 1090\,\mathrm{kg\,m^{-3}}$, and liquid occupancy $f_o = 0.90$. HRT denotes hydraulic retention time. Total digester volume differs from daily gas output.

$$\boxed{\begin{gathered} G = N_p g_c + N_L t_L g_L, \quad G_{\text{cow}} = m_{\text{dung,cow}} f_c f_d Y, \quad N_{\text{cow}} = \left\lceil \frac{G}{G_{\text{cow}}} \right\rceil, \\ m_d = \frac{G}{Y}, \quad m_w = \frac{m_d}{f_d}, \quad V_{\text{dig}} = \frac{(m_w + m_{\text{water}})\,\text{HRT}}{\rho_s f_o}. \end{gathered}} \tag{19.1}$$

G is daily gas demand; f_c is collectible fraction; $f_d = 0.18$ is dry-matter fraction; Y is gas yield per kg of dry matter; m_w is fresh-dung feed.

1. Daily cooking and lighting demand.

$$\begin{aligned} G_c &= 5(0.227) = 1.135\,\mathrm{m^3\,day^{-1}}, \\ G_L &= 2(3)(0.126) = 0.756\,\mathrm{m^3\,day^{-1}}, \\ G &= 1.135 + 0.756 = 1.891\,\mathrm{m^3\,day^{-1}}. \end{aligned}$$

2. Gas yield per cow and required herd.

$$\begin{aligned} m_{w,\mathrm{cow}} &= 10(0.7) = 7.000\,\mathrm{kg\,day^{-1}}, \\ m_{d,\mathrm{cow}} &= 7.000(0.18) = 1.260\,\mathrm{kg\,day^{-1}}, \\ G_{\mathrm{cow}} &= 1.260(0.34) = 0.4284\,\mathrm{m^3\,day^{-1}}, \\ N_{\mathrm{cow}} &= \left\lceil \frac{1.891}{0.4284} \right\rceil = \lceil 4.4141 \rceil = 5. \end{aligned}$$

3. Minimum fresh-dung and slurry feed.

$$\begin{aligned} m_d &= \frac{1.891}{0.34} = 5.5618\,\mathrm{kg\,day^{-1}}, \\ m_w &= \frac{5.5618}{0.18} = 30.8987\,\mathrm{kg\,day^{-1}}, \\ m_s &= m_w + m_{\mathrm{water}} = 2(30.8987) = 61.7974\,\mathrm{kg\,day^{-1}}. \end{aligned}$$

4. Demand-based digester volume.

$$\begin{aligned} \dot{V}_s &= \frac{61.7974}{1090} = 0.056695\,\mathrm{m^3\,day^{-1}}, \\ V_{\mathrm{dig}} &= \frac{(0.056695)(50)}{0.90} = 3.1497\,\mathrm{m^3}. \end{aligned}$$

5. Volume if all collectible dung from the rounded herd is fed.

$$\begin{aligned} m_{w,\mathrm{herd}} &= 5(10)(0.7) = 35.000\,\mathrm{kg\,day^{-1}}, \\ V_{\mathrm{herd}} &= \frac{2(35.000)(50)}{1090(0.90)} = 3.5678\,\mathrm{m^3}. \end{aligned}$$

Final answer. Required herd: 5 cows. Minimum demand-based volume: $3.15\,\mathrm{m^3}$. For all collectible dung from the 5 cows, provide $3.57\,\mathrm{m^3}$.

Assumption / data note. The heating value is not required for the requested sizing. Without added dilution water, the computed volumes would be half these values; that is a different feed assumption.

20 Sunset angle and day length at 33 degrees north

01 December 2022 Q3(b) 10 marks Source PDF p. 19

Original question.

Determine the sunset hour angle and day length at a location latitude of 33° on March 29. The standard longitude for IST is 81° and 44′ E.

Worked solution

1. Day number and latitude.

March 29 in a non-leap year gives $n = 31 + 28 + 29 = 88$. Thus $n = 88$ and $\phi = 33°$.

$$\boxed{\delta = 23.45° \sin\left[\frac{360°(284+n)}{365}\right], \qquad \omega_s = \cos^{-1}(-\tan\phi\tan\delta), \qquad N = \frac{2\omega_s}{15} \quad \text{hours.}} \tag{20.1}$$

Here δ is declination and ω_s is the positive sunset hour angle. All trigonometric calculations use degree mode.

2. Declination.

$$\begin{aligned}\delta &= 23.45° \sin\left[\frac{360°(284+88)}{365}\right] \\ &= 2.8189°.\end{aligned}$$

3. Sunset hour angle.

$$\begin{aligned}\cos\omega_s &= -\tan(33°)\tan(2.8189°) = -0.031976, \\ \omega_s &= \cos^{-1}(-0.031976) = 91.8324°.\end{aligned}$$

4. Daylight duration.

$$\begin{aligned}N &= \frac{2(91.8324)}{15} = 12.2443\,\text{h} \\ &= 12\,\text{h}\ 14.66\,\text{min}.\end{aligned}$$

Final answer. $\omega_s = 91.83°$, $N = 12.244\,\text{h}$ (approximately 12 h 15 min).

Assumption / data note. No year is given, so a non-leap year is assumed. The printed standard longitude is unnecessary for the requested hour angle and day length. Geometric daylight is used.

21 Maximum possible torque: 125 m rotor

01 December 2022 Q5(a) 10 marks Source PDF p. 20

Original question.

A HAWT has following data:

Speed of wind = 15 m/s at 1 atm and 15 °C
Diameter of rotor = 125 m
Speed of rotor = 45 rpm

Calculate the maximum possible torque produced at the shaft.

Worked solution

Assumption / data note. Maximum possible shaft output is interpreted as the ideal Betz-limit aerodynamic output, with no unspecified drivetrain loss. Actual turbine torque would be lower.

$$\boxed{\begin{aligned} &\rho = \frac{p}{R_{\text{air}}T}, \quad A = \frac{\pi D^2}{4}, \quad P_w = \frac{1}{2}\rho A V^3, \\ &P_{\max} = \frac{16}{27}P_w, \qquad \Omega = \frac{2\pi N}{60}, \qquad \tau_{\max} = \frac{P_{\max}}{\Omega}. \end{aligned}} \tag{21.1}$$

1. Air density at the stated conditions.

Use $p = 1\,\text{atm} = 101\,325\,\text{Pa}$ and $R_{\text{air}} = 287\,\text{J kg}^{-1}\,\text{K}^{-1}$.

$$\begin{aligned} T &= 15 + 273.15 = 288.15\,\text{K}, \\ \rho &= \frac{101\,325}{287(288.15)} = 1.225226\,\text{kg m}^{-3}. \end{aligned}$$

2. Swept area and power in the wind.

$$\begin{aligned} A &= \frac{\pi(125)^2}{4} = 12\,271.846\,\text{m}^2, \\ P_w &= \tfrac{1}{2}(1.225226)(12271.846)(15)^3 \\ &= 25.372881\,\text{MW}. \end{aligned}$$

3. Betz-limit output and angular speed.

$$\begin{aligned} P_{\max} &= \frac{16}{27}(25.372881) = 15.035781\,\text{MW}, \\ \Omega &= \frac{2\pi(45)}{60} = 4.712389\,\text{rad s}^{-1}. \end{aligned}$$

4. Maximum shaft torque.

$$\begin{aligned} \tau_{\max} &= \frac{15.035781 \times 10^6}{4.712389} \\ &= 3\,190\,691.88\,\text{N m}. \end{aligned}$$

Final answer. $\tau_{\max} = 3.1907\,\text{MN m} = 3\,190.69\,\text{kN m}$.

Assumption / data note. Use watts and radians per second in $\tau = P/\Omega$. Rotor rpm cannot be inserted directly as angular speed.

Results checklist

The assumptions and qualifications in each worked solution remain part of its answer.

No.	Final result
01	92.26°; 12.302 h
02	19.079, 9.769, 8.303 MW
03	$\theta = 30.50°$
04	Rs. -4 632.59; Rs. -6 191.61
05	10.250 MW; 5.248 MW; 401.904 kN
06	102.34°; 13.645 h
07	21.241 m s^{-1}; $a = 0.25$; $C_p = 0.5625$; powers depend on geometry
08	12 cows; 8.72 m^3 minimum; 8.81 m^3 full herd
09	Rs. 8 880.70; Rs. 9 392.33
10	$\theta = 17.18°$
11	Rs. 2 240.13; Rs. 2 227.14
12	8.713 MW; 3.777 MW; 314.568 kN
13	92.80°; 12.374 h
14	1.9604 MN m
15	13.59 m^3; 2.396 kW gross; 1.677 kW useful
16	Rs. 1 953.72; Rs. 2 830.79
17	49.20° printed meridian; 48.77° actual IST meridian
18	Rs. -114 993.72; Rs. 148 513.07
19	5 cows; 3.15 m^3 minimum; 3.57 m^3 full herd
20	91.83°; 12.244 h
21	3.1907 MN m

Calculation references

These references support methods and stated assumptions; examination data come from the supplied papers.

1. NASA, *Evaluation of Surface Energy and Radiation*: conventional solar declination approximation. NASA report.
2. NOAA, *General Solar Position Calculations*: solar-time and zenith geometry. The equation-of-time approximation used here is stated explicitly in Problem 17. NOAA equations.
3. QBlade, *Wind: actuator-disc theory*; MIT OpenCourseWare, *Wind Power Fundamentals*: induction, thrust, extracted power and Betz limit. QBlade theory; MIT lecture.
4. Government of India, biogas guidance: 1:1 cow-dung-to-water mixing. Biogas guidance. India Meteorological Department, *Solar Radiant Energy Over India*: standard longitude 82.5° E. IMD reference.